Fundamentals of Cybersecurity

Mohammad Hosseini

Faculty of Computer Science and Engineering
Shahid Beheshti University

Network Security Countermeasures 2

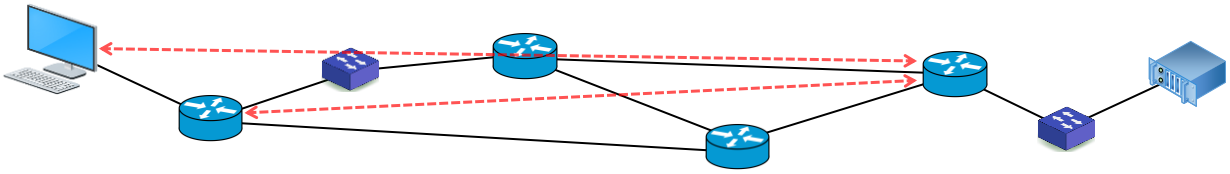


Recap

- End-to-end encryption
  - e.g., TLS
- Link encryption
  - e.g., WiFi

IPSec

- The IP security protocol, more commonly known as IPsec, provides security at the network layer.
- IPsec secures IP datagrams **between any two network-layer entities**, including hosts and routers.
- With network-layer confidentiality between a pair of network entities, the sending entity **encrypts** the payloads of all the datagrams it sends to the receiving entity.
  - The datagram can be a TCP segment, a UDP segment, an ICMP message, and so on.



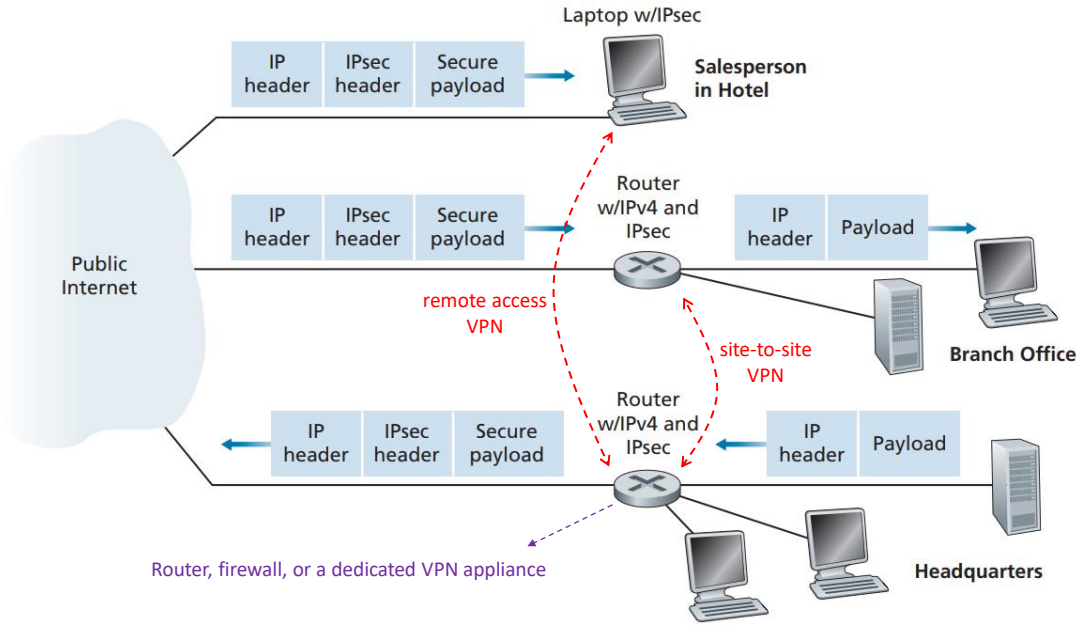
- IPSec also provides authentication and data integrity.

Virtual Private Network (VPN)

An institution that extends over multiple geographical regions often desires its own IP network, so that its hosts and servers can send data to each other in a secure and confidential manner.

- Solution 1: deploying a stand-alone physical network (including routers and links) that is completely separate from the public Internet. (or using leased-lines)
  - Such a disjoint network, dedicated to a particular institution, is called a private network.
  - Very costly
- Solution 2: creating VPNs (overlay networks) over the existing public Internet.
  - To provide confidentiality, the inter-office traffic is encrypted before it enters the public Internet.

Virtual Private Network (VPN)



Virtual Private Network (VPN)

- A virtual private network simulates the security of a dedicated, protected communication line on a shared network.
- When two of the institution's hosts communicate over a path that traverses the public Internet, the traffic is encrypted before it enters the Internet.
- The gateway router in headquarters converts the ordinary IPv4 datagram into an IPsec datagram.
- This IPsec datagram has a traditional IPv4 header, so that the routers in the public Internet process the datagram as if it were an ordinary IPv4 datagram.
- When the IPsec datagram arrives at the salesperson's laptop, the OS in the laptop decrypts the payload and passes the unencrypted payload to the upper-layer protocol (for example, to TCP or UDP).

IPSec

In the IPsec protocol suite, there are two principal protocols.
(In an IPSec-based connection, one of these two is used)

- Authentication Header (AH) protocol
  - Provides source **authentication** and **data integrity** but does not provide **confidentiality**.
- Encapsulation Security Payload (ESP) protocol
  - Provides source **authentication**, **data integrity**, and **confidentiality**.

IPSec connection

Before sending IPsec datagrams, the source and destination entities create a **network-layer logical connection**.

Security Association (SA)

- SA is simplex → it is unidirectional from source to destination.
  - If secure traffic is needed in both directions, two SAs need to be established, one in each direction.
- A host (such as a server or a firewall) might have several security associations in effect for concurrent communications with different remote clients.
  - SAs are stored in a database called the Security Association Database (**SAD**).

IPSec connection

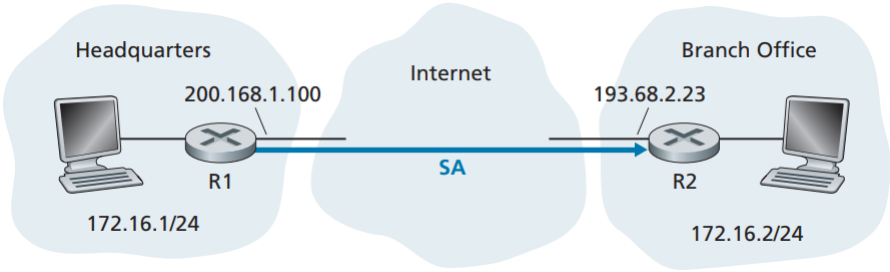
- Not all traffic sent into the Internet by the gateway routers or by the laptops will be IPsec secured.
  - IPsec entity maintains a data structure called the **Security Policy Database** (SPD).
  - The SPD indicates what types of datagrams (as a function of source IP address, destination IP address, and protocol type) are to be IPsec processed, and which SA should be used.

SA state

Gateway router maintains state information about each SA:

- A 32-bit identifier for the SA, called the **Security Parameter Index** (SPI)
- The origin interface of the SA and the destination interface of the SA (IP addresses)
- The type of encryption to be used (e.g., 3DES with CBC)
- The encryption key (and parameters such as the initialization vector)
- The type of integrity check (e.g., HMAC with MD5)
- The authentication key

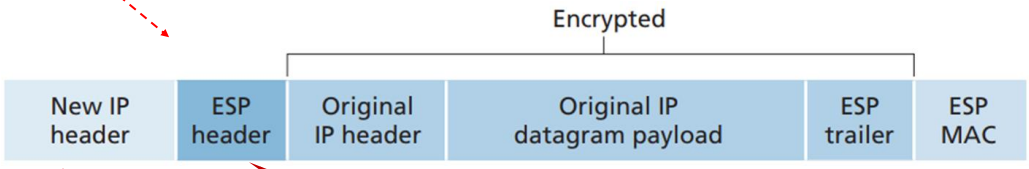
IPSec example



IPsec datagram

IPsec has two different packet forms:

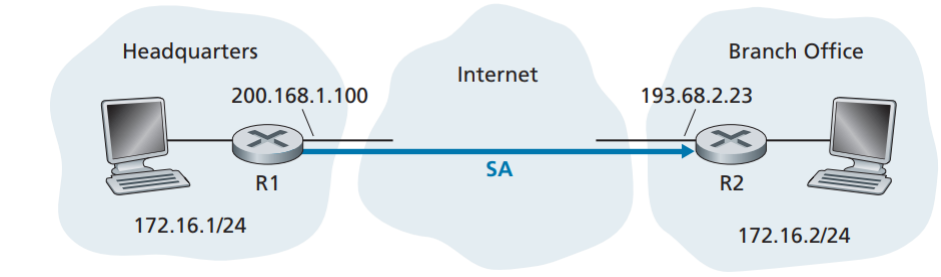
- Transport mode
- Tunnel mode (we focus on this mode, which is more widely deployed)

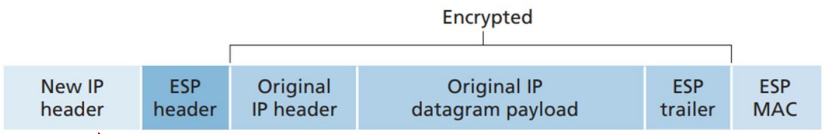


IP addresses of the origin and destination gateway interfaces

Contains **SPI**, which is used to look up keys and other relevant information when a secure packet arrives.

IPsec datagram

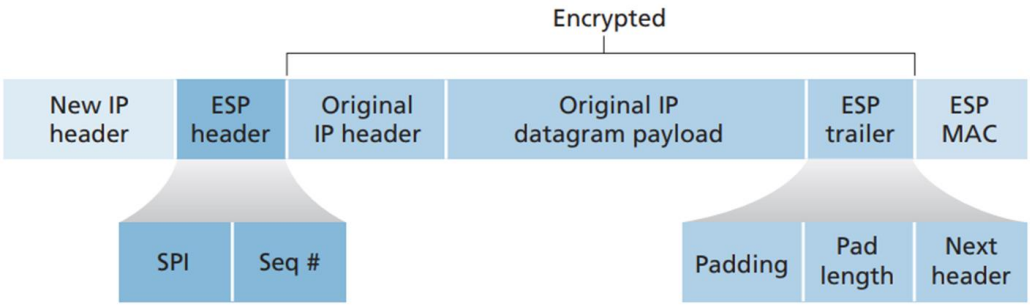




Source IP: `200.168.1.100`
Destination IP: `193.68.2.23`
Protocol number: `50` (IPSec)

Source IP: `172.16.1.17`
Destination IP: `172.16.2.48`

IPsec datagram



Man-in-the-middle

- The protocol number, the source IP address, the destination IP address, and data in the original datagram are hidden from a man-in-the-middle.
  - An intruder does not know if the packet is carrying TCP, UDP, HTTP, SMTP, or any other type of application data.
  - This confidentiality thus goes a lot farther than SSL.
- Flipping some bits or masquerading a gateway router can be easily detected by MAC integrity check.
- Replay attacks can also be detected by sequence numbers.

Key Management in IPsec

- When a VPN has a small number of end points, the network administrator can manually enter the SA information (encryption/authentication algorithms and keys, and the SPIs) into the endpoints.
- Large, geographically distributed deployments require an automated mechanism for creating the SAs.
  - IPsec does this with the Internet Key Exchange (IKE) protocol.
  - IKE has some similarities with the handshake in SSL.

Remote access VPN

- The VPN server assigns a **private virtual IP address** to the client.
  - The private IP address is configured on the client during the VPN connection setup process.
  - The client sends packets with the private virtual IP address in the inner IP header.
  - The private virtual IP address is used for communication between the client and the internal resources of the organization (e.g., file servers, databases).
  - This address is only valid inside the VPN tunnel and is not visible on the public Internet.
- The VPN client modifies the host's routing table to send specific traffic through the tunnel.

How is a VPN utilized to conceal identity, ensure anonymity, bypass geo-restrictions, or avoid censorship?

Other VPN protocols

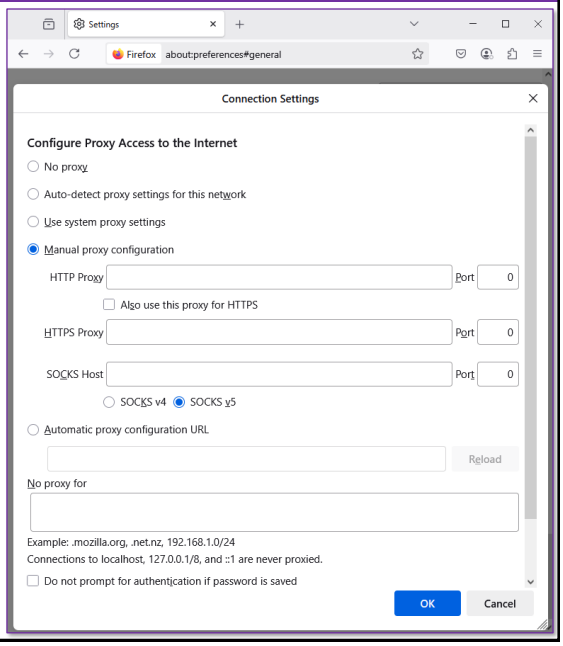
- PPTP
  - Developed by Microsoft in mid-1990s.
  - Considered outdated and insecure.
  - PPTP has many well-known security issues.
- OpenVPN
  - Uses SSL/TLS
- WireGuard
  - High-performance VPN

Proxy

- A proxy server is a server that acts as an intermediary between a client requesting a resource and the server providing that resource.
- Operates at the application layer.



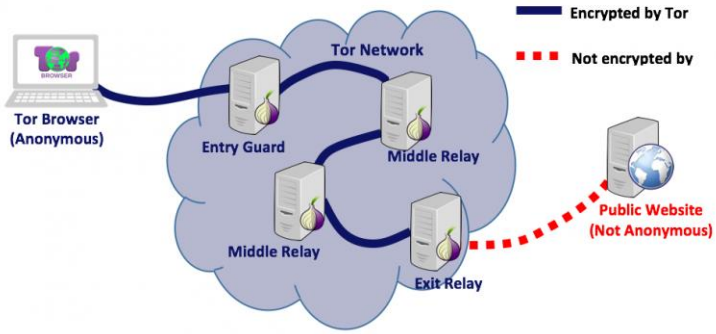
- e.g., HTTP proxy, SOCKS



How do proxies such as *v2ray* or *hiddify* work?

Onion routing

- Onion routing is a technique where data is sent through a series of intermediate hosts (called “relays”) with multiple layers of encryption.
- The data is encrypted multiple times (at the client), with each layer of encryption corresponding to a specific relay in the network.
- Each relay can only decrypt its own layer of encryption, revealing the next destination.



- TOR is a free and open-source implementation of onion routing, which provides a network of volunteer-operated relays.

How does *shecan* work?